MULT20015: Tutorial 12

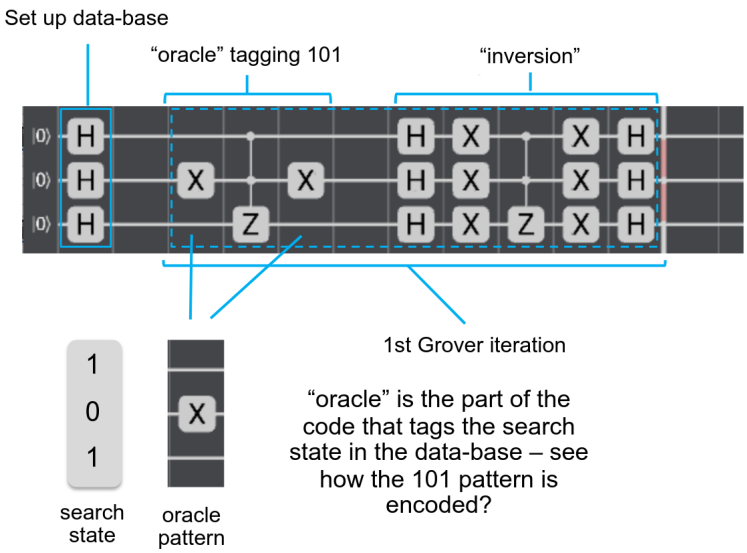
Welcome to Practice Class-12 of MULT20015 Elements of Quantum Computing, covering exercises relevant to the material presented in lectures in Week 12

The purpose of this week's tutorial is to

- See how quantum computers work in practice
- Access the IBM Quantum systems and get a feel for how they work
- Compare ideal circuit outputs from QUI with that from real quantum devices

A demonstration of Grover's Algorithm on IBM devices

Your tutor will walk you through how quantum circuits on IBM are programmed using the following circuit as an example.



IBM Quantum devices are (typically) programmed using a Software Development Kit (SDK) called Qiskit programmed in Python. This demonstration uses Qiskit so that you can get a feel for how Quantum Computing researchers actually create their circuits. Later on in this tutorial, you will use a GUI tool written by IBM to program circuits to run on quantum devices.

Exercise 1 After the tutor has shown you the results of Grover's algorithm on the IBM device, program the following circuit into QUI and compare the output to that of the IBM device. What is the fidelity of the algorithm run on the IBM device compared to the ideal output from QUI? Can you explain the differences between the two?

Exercise 2 The QUI allows for an error model to be applied with user-defined parameters (e.g. error rate/probability). Modify your QUI circuit to implement a continuous error model, and by experimenting with the error parameters provide an estimate of the value that gives results commensurate with those obtained on the physical IBMQ device.

Exercise 3 If we were to run more Grover iterations on the IBM device do you expect the fidelity to increase, decrease or stay roughly the same? Discuss with your classmates and tutor.

Exercise 4 On QUI program another iteration of Grover with the same error model applied, determine the fidelity of the simulation and write down this result somewhere for reference. We will refer back to this result in Exercise 7.

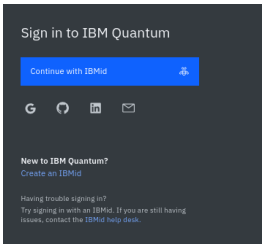
Creating an IBM account and submitting circuits

Now it's your turn to submit circuits to the IBM quantum devices. You may not see the output before the end of the tutorial, but you should get a feel for the workflow. Work through the following to create an IBM account and submit your first circuit to a quantum device!

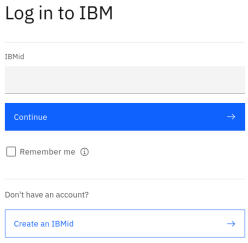
Exercise 5

1. Visit the following website `https://quantum.ibm.com/` and click on "Create an IBMid" (twice!) to sign-up

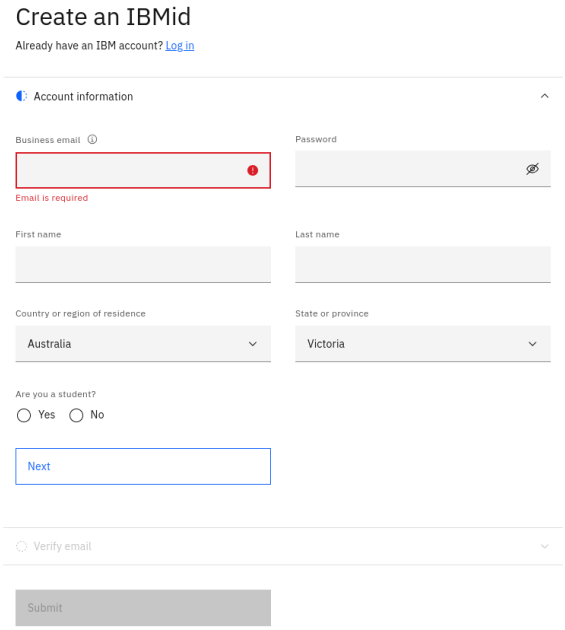
Log-in page



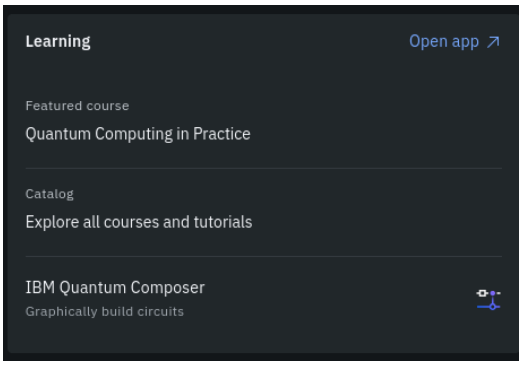
First "Create an IBMid" page



2. Fill in your details to create an account. If you're unsure of anything, ask your tutor for help.



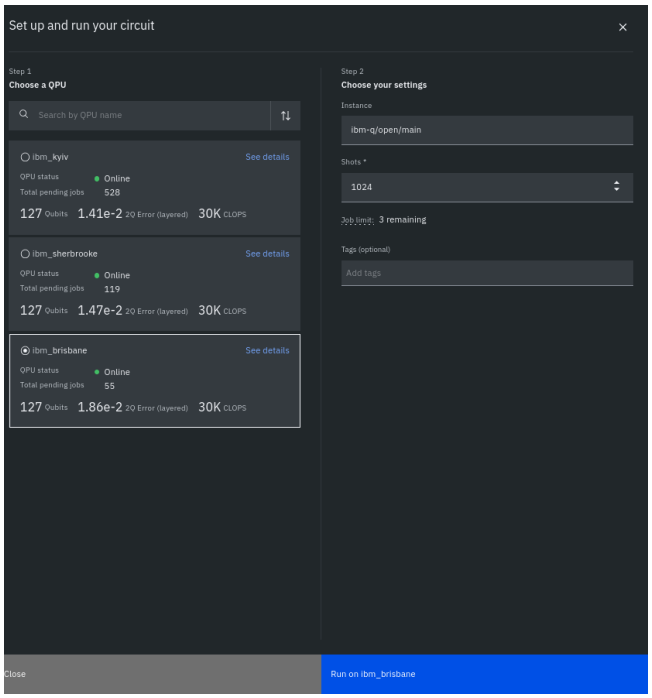
3. After signing up, on the home page click on "IBM Quantum Composer" found near the bottom of the page.



4. Using the IBM Quantum Composer, construct a Grover circuit with two iterations using the oracle from the tutor demonstration in the first section. You may add gates to the circuit by dragging and dropping gates from the operations panel on the left. N.B. You will need to use H gates to turn Toffoli gates into CCZ gates.

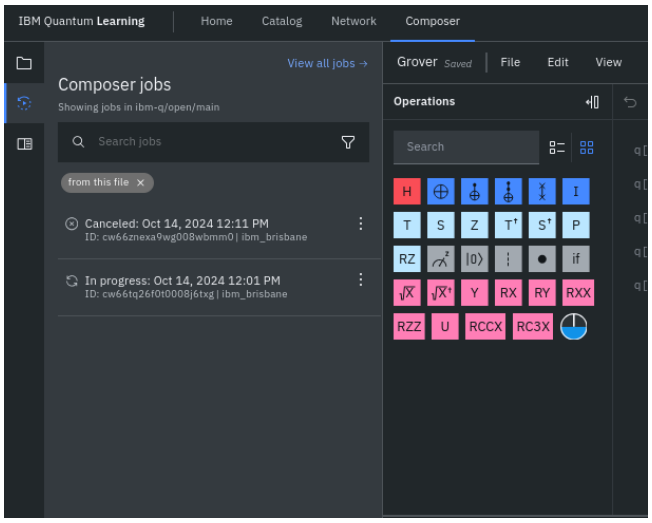
5. Once you have finished constructing your circuit, you will need to add measurement gates to the end as we will be running this circuit on a quantum device.

6. Click on "Setup and run" in the top right hand side. A pop-up similar to this will appear. You can click



"See details" to see the error rates and topology of the various devices available to you. After exploring for a bit select a device and specify a number of shots (the default 1024 should be fine) then click "Run on..." to submit the job.

7. To see the list of jobs for this circuit click on the "Composer Jobs" found on the left hand side.



8. Select the job you're interested in and then "See more details" to see the circuit that is run on the device (that is the transpiled circuit) and job results (once the job has completed).

Exercise 6 You will see that the transpiled circuit looks quite different to the circuit you constructed. This is because the gates available on the quantum device are different to that which we typically use to describe quantum algorithms. Are you able to determine which combination of "native" device gates correspond to the gates you used to construct Grover's algorithm?

Exercise 7 In your own time after class, check back in on the IBM Quantum website to see if your job was completed successfully. Is the fidelity of the circuit consistent with your answer to Exercise 3? Is the fidelity similar to the noisy QUI circuit you ran in class?

You should now hopefully have an idea on how quantum circuits are run on real devices. If you're keen you may want to run more complicated circuits see if you can estimate the capability of existing quantum hardware by running increasingly more complex circuits.